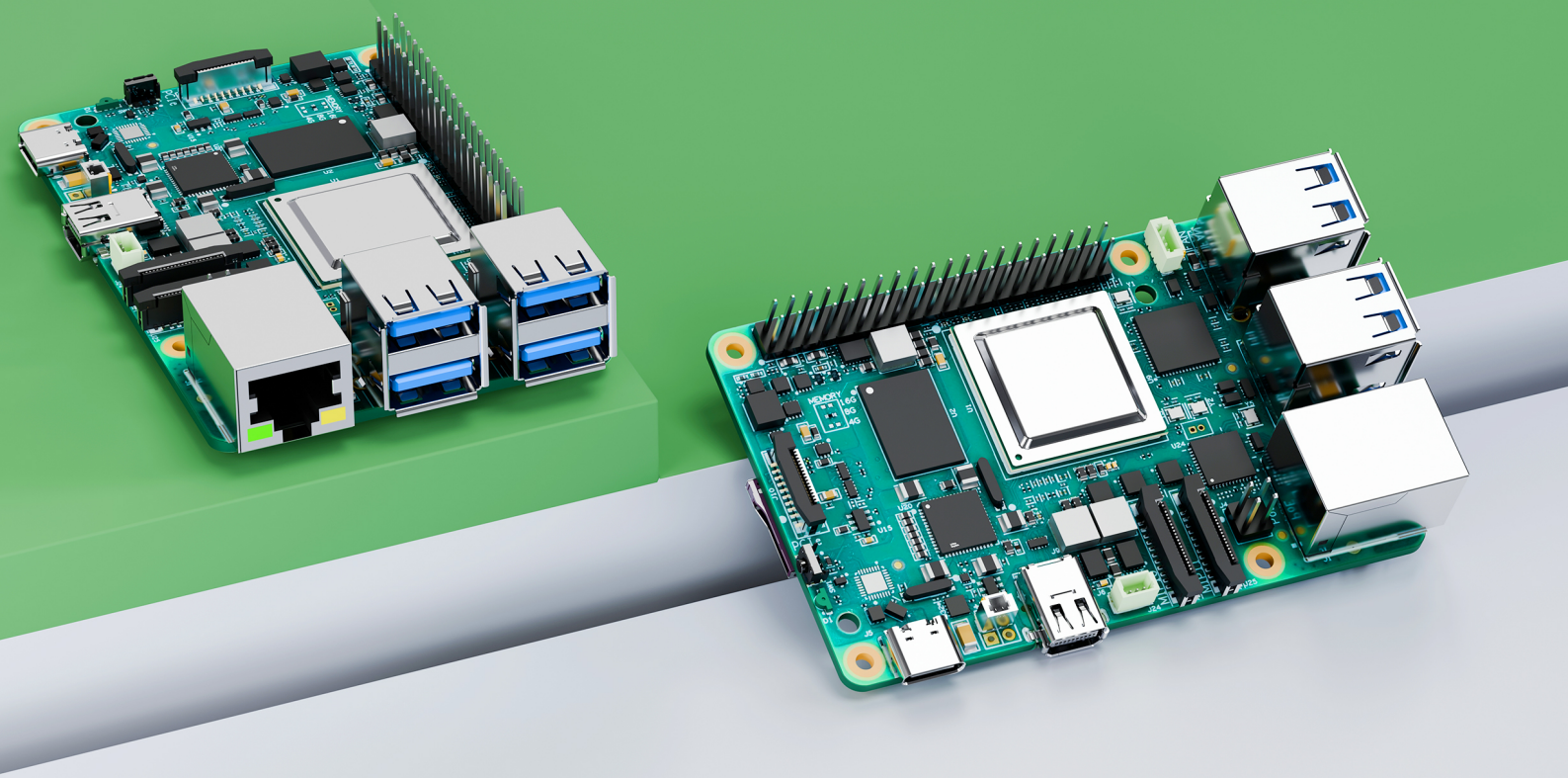
InHand Networks

Embrace Edge AI, Empower Intelligent Vision



# Mo 68A AI Single Board Computer

· Open SDK

· Linux Distribution

· HAT Expansion

· Edge AI

# 1. Product Overview

**The Mo 68A is an 8 TOPS edge AI single-board computer based on the TI AM68A SoC, with a form factor compatible with standard SBCs, designed for AI vision boxes, intelligent cameras, and edge AI inference terminals.**

**Product Features:**

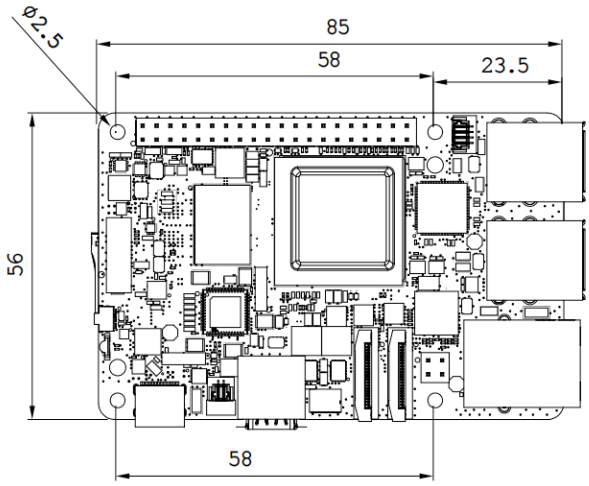
- **Edge AI:** 8 TOPS on-device AI acceleration, TI TIDL supports TFLite / ONNX
- **Vision Processing:** Dual 4-lane MIPI CSI-2, VPAC / DMPAC, 4K@60fps codec
- **Rich Interfaces:** Gigabit Ethernet, PCIe 3.0, 4 × USB 3.0, mini DP
- **SBC Compatibility:** Standard form factor, 40-pin HAT compatibility, support for mainstream enclosures and accessories
- **High Security:** Secure Boot, TrustZone, OP-TEE, hardware crypto acceleration

## Core Technical Specifications

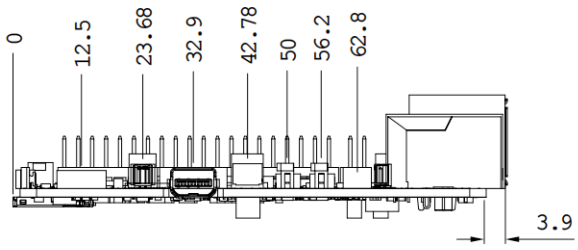
| Technical Indicator | Specification |
|---|---|
| AI Inference | 8 TOPS, TI TIDL supports TFLite / ONNX models |
| Vision Processing | Dual 4-lane MIPI CSI-2 RX, VPAC / DMPAC, 4K@60fps H.265 / H.264 hardware codec |
| Operating System | Debian 13 Trixie, Linux 6.12 |
| Network | 1 × Gigabit Ethernet |
| Security | Secure Boot, TrustZone, OP-TEE, AES-256 |
| Development | Python / C / C++, OpenCV, Open SDK |
| CPU | 2 × Cortex-A72 @ 2.0 GHz |
| Memory | LPDDR4 4 GB / 8 GB (default) |
| Storage | Micro SD |

| Technical Indicator | Specification |
| --- | --- |
| Power | USB Type-C 5 V / 5 A DC, 25 W (MAX) |
| Dimensions | 85 × 56 mm |
| Operating Temperature | 0 ~ 50 °C |

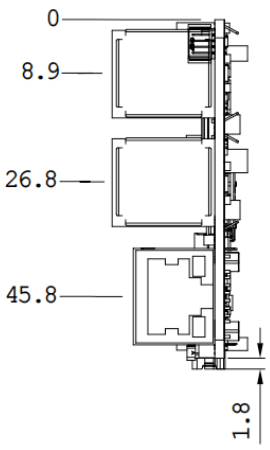
# 2. Product Dimensions




Front View




Side View




Interface View

Note:

1. All dimensions are in millimeters (mm).
2. All dimensions are approximate values for **reference only**.
3. The dimensions shown **shall not be used for production or processing**.
4. Dimensions shall comply with part and manufacturing tolerance requirements.
5. Dimensions are subject to change without notice.

# 3. Hardware Specifications

| Category / Parameter | Specification |
|---|---|
| **Hardware Platform** | |
| CPU | TI AM68A, 2 × Cortex-A72 @ 2.0 GHz |
| AI Accelerator | 2 × C7x DSP + Deep Learning Accelerator, 8 TOPS |
| ISP / Vision | On-chip ISP + VPAC (RGB-IR, WDR, LDC) |
| RAM | LPDDR4 4 GB / 8 GB (default) |
| **Interfaces** | |
| Ethernet | 1 × Gigabit Ethernet |
| USB | 4 × USB 3.0 Type-A |
| PCIe | 1 × PCIe 3.0 |
| Display | 1 × mini DP, up to 2 × 4-lane MIPI DSI |
| Camera | up to 2 × 4-lane MIPI CSI-2 |
| Audio | I2S via 40-pin connector |
| 40-pin Connector | GPIO / I2C / I2S / SPI / UART / PCM, HAT-compatible |
| Fan Connector | 1 × 4-pin fan connector (5 V, PWM, GND, TACH) |
| Button | 1 × Reset button |
| Storage | Micro SD |
| Debug UART | 1 × TTL UART |
| LED | PWR, STATUS |
| **Security** | |

| Category / Parameter | Specification |
|---|---|
| Secure Boot | Support (ARM TrustZone + OP-TEE) |
| **Power** | |
| Power Input | USB Type-C 5 V / 5 A DC |
| Power Consumption | 25 W (MAX) |
| **Mechanical** | |
| Dimensions | 85 × 56 mm |
| Weight | 53 g |
| Housing | PCB |
| Cooling | Active fan (optional) |
| **Others** | |
| RTC | Support (battery backup) |
| Watchdog | Hardware watchdog supported |
| **Environmental** | |
| Operating Temperature | 0 ~ 50 °C |
| Storage Temperature | -20 ~ 70 °C |

# 4. Software Specifications

| Category / Parameter | Specification |
|---|---|
| **Operating System** | |
| OS | Debian 13 Trixie |

| Category / Parameter | Specification |
|---|---|
| Kernel | Linux Kernel 6.12 |
| **AI & Vision** | |
| AI Runtime | TI TIDL, supports TFLite / ONNX |
| Vision SDK | TI EdgeAI SDK |
| Camera Framework | V4L2 |
| Display Framework | DRM / KMS |
| **Network Features** | |
| IP Application | TCP / UDP, ICMP, DNS, DHCP |
| IP Routing | Static routing |
| **Security** | |
| Secure Boot | Support |
| TrustZone | Support |
| OP-TEE | Support |
| Crypto Accelerator | Hardware AES-256 |
| **Development** | |
| Languages | Python, C / C++ |
| Libraries | OpenCV, GStreamer, NumPy |
| Package Manager | apt (Debian) |
| Open SDK | Supports custom system build by customer |
| **System Management** | |
| Remote Access | SSH |

| Category / Parameter | Specification |
| --- | --- |
| Firmware Upgrade | SD card flash |
| Debug | UART console |

# 5. Ordering Information

## Model Code Rule

**Model code:** Mo68A-<x>-<y>

Mo68A: Base Model

<-x>: Memory

<-y>: MIPI configuration

## Product Models

| Base Model | -x | Memory | -y | MIPI Configuration |
|---|---|---|---|---|
| Mo68A | 4G | 4 GB | DC | DSI on MIPI0, CSI on MIPI1 |
| Mo68A | 8G | 8 GB | DC | DSI on MIPI0, CSI on MIPI1 |

**Example**
Model: Mo68A-8G-DC
"Mo68A" means Base Model.
"-8G" means 8 GB memory.
"-DC" means DSI on MIPI0, CSI on MIPI1.

# 6. Contact Us

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